

Topics for week 2

Chapter 1: Linear Systems and Matrix Algebra (Cont.)

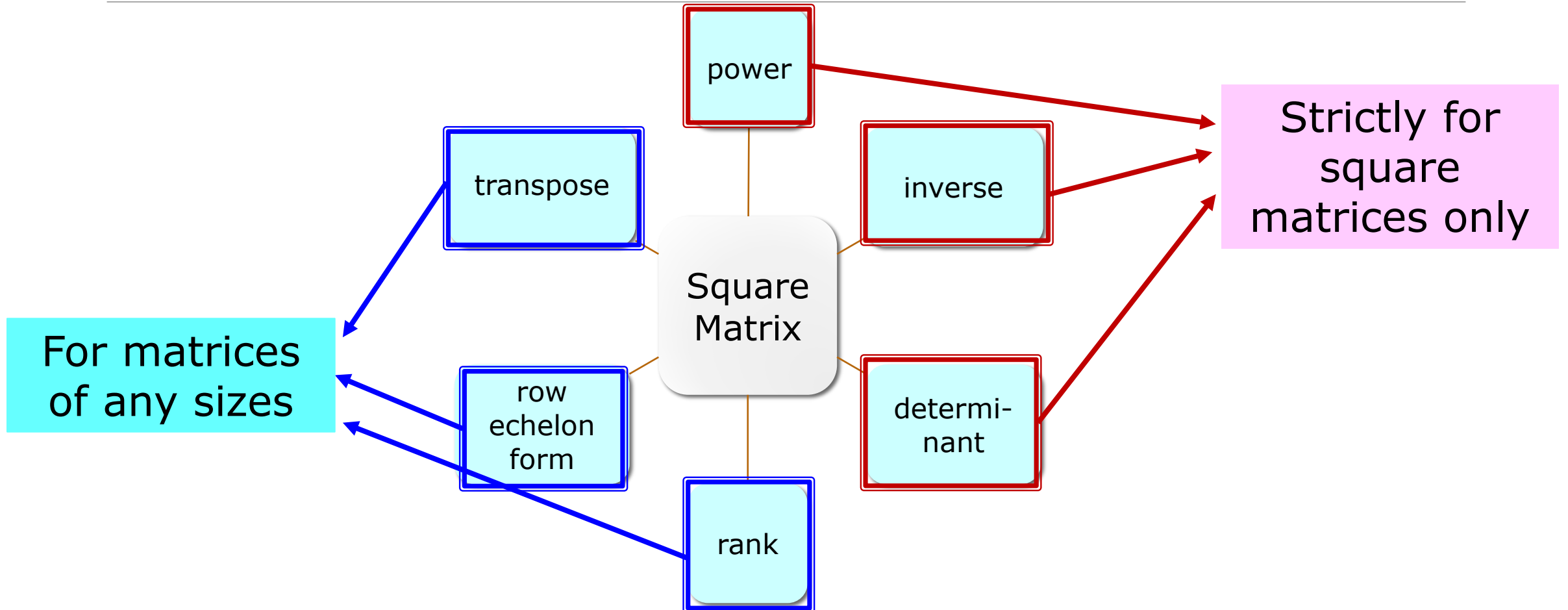
- 1.6 Matrix inverses 2
- 1.7 Determinants of matrices
- 1.8 Ranks of matrices

Chapter 2: Vector Spaces

- 2.1 Vectors in n -space
- 2.2 Linear combination and linear span

Square Matrices

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix}$$



Finding Matrix Inverse

We can find the inverse only when the matrix is **non-singular**

For 2 x 2 matrix $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \longrightarrow \mathbf{A}^{-1} = \begin{pmatrix} \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{pmatrix}$ provided $ad - bc \neq 0$

For n x n matrix $\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix}$

Example

$\mathbf{A} = \left(\begin{array}{ccc|ccc} 2 & 0 & -1 & 1 & 0 & 0 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \end{array} \right)$ **(A | I)**

Gauss Jordan Elimination

Make sure this is **I**

$\mathbf{A}^{-1}$

$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 2/3 & -1/3 & 1/3 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1/3 & -2/3 & 2/3 \end{array} \right)$

reduced row echelon form of **(A | I)**

Determine singularity of a matrix

Two methods

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix}$$

singular

RREF $\neq \mathbf{I}$

det = 0

non-singular

RREF = $\mathbf{I}$

det $\neq 0$

2 x 2 and 3 x 3 Determinant

For a 2 x 2 matrix, $\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

$$\det \mathbf{A} = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

For a 3 x 3 matrix

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Inductive method

Expand $n \times n$ determinant into $(n-1) \times (n-1)$ determinants

Finding Determinant

- Inductive method (Expand into smaller sized determinants)
- Gaussian elimination (for larger matrices)
- Special cases
 - triangular/diagonal matrices Multiply the diagonal entries
 - identical rows/columns $\det = 0$
 - zero rows/columns $\det = 0$

n x n Determinant

Interchanging two rows
will change a determinant
to its "negative"

For n x n matrix (use Gaussian elimination)

$$\begin{vmatrix} 3 & -1 & 1 & 1 \\ 3 & -1 & 2 & 1 \\ 0 & 2 & 4 & 0 \\ 0 & 0 & 2 & -1 \end{vmatrix} \xrightarrow{R_2 - R_1} \begin{vmatrix} 3 & -1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 2 & 4 & 0 \\ 0 & 0 & 2 & -1 \end{vmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{vmatrix} 3 & -1 & 1 & 1 \\ 0 & 2 & 4 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & -1 \end{vmatrix}$$

$$\begin{vmatrix} 3 & -1 & 1 & 1 \\ 0 & 2 & 4 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{vmatrix} \xrightarrow{R_4 - 2R_3} \begin{vmatrix} 3 & -1 & 1 & 1 \\ 0 & 2 & 4 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{vmatrix}$$

(upper triangular)

You need to keep track
of the steps

$$= -(3)(2)(1)(-1) = 6$$

Rank of a Matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 2 & 1 \\ 3 & 6 & 6 & 3 \\ 4 & 9 & 9 & 5 \\ -2 & -1 & -1 & 1 \\ 5 & 8 & 9 & 4 \\ 4 & 2 & 7 & 3 \end{pmatrix} \xrightarrow[\text{Elimination}]{\text{Gaussian}} \begin{pmatrix} 1 & 2 & 2 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$\text{rank}(\mathbf{A}) = 3$

Non-zero rows

Pivot Columns

- No. of non-zero rows in REF of $\mathbf{A}$
- No. of leading entries in REF of $\mathbf{A}$
- No. of pivot columns in REF of $\mathbf{A}$

They are all equal

Non-Singular Matrices

$n \times n$ square matrix $\mathbf{A}$

$\mathbf{A}$ non-singular matrix

$\mathbf{A}$ has an inverse $\mathbf{A}^{-1}$

row echelon form of $\mathbf{A}$ does not have zero row

reduced row echelon form of $\mathbf{A}$ is the identity matrix

homogeneous system $\mathbf{Ax} = \mathbf{0}$ has only trivial solution

non-homogeneous system $\mathbf{Ax} = \mathbf{b}$
has exactly one solution

$\det \mathbf{A} \neq 0$

Full rank means $\text{rank } \mathbf{A} = n$

$\mathbf{A}$ is full rank

$\mathbf{A}$ singular matrix

$\mathbf{A}$ does not have an inverse $\mathbf{A}^{-1}$

row echelon form of $\mathbf{A}$ has a zero row

reduced row echelon form of $\mathbf{A}$ is not the identity matrix

homogeneous system $\mathbf{Ax} = \mathbf{0}$ has non-trivial solutions
in addition to the trivial solution

non-homogeneous system $\mathbf{Ax} = \mathbf{b}$
has either no solution or infinitely many solutions

$\det \mathbf{A} = 0$

$\mathbf{A}$ is not full rank

Not full rank means $\text{rank } \mathbf{A} < n$

Example 1

Suppose $\mathbf{A}$ is a 3×3 matrix
and the system $\mathbf{Ax} = \mathbf{b}$ has no solution

Are the following true?

- i. $\det \mathbf{A} = 0$
- ii. $\text{rank } \mathbf{A}$ cannot be 3
- iii. the homogeneous system $\mathbf{Ax} = \mathbf{0}$ has non-trivial solutions
- iv. $\mathbf{A}$ has a zero row

Example 2

$$\mathbf{A} = \begin{pmatrix} 7 & 8 & \square \\ \square & 9 & 6 \\ 4 & \square & 3 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 1 & 2 & 3 \end{pmatrix} \quad \begin{array}{l} \text{two identical rows} \\ \det \mathbf{B} = 0 \end{array}$$

How many solutions does the system $\mathbf{ABx} = \mathbf{0}$ have?

Ans: Infinitely many

$$\det \mathbf{AB} = \det \mathbf{A} \times \det \mathbf{B} = 0$$

So $\mathbf{AB}$ is singular

So $\mathbf{ABx} = \mathbf{0}$ has non-trivial solutions

So $\mathbf{ABx} = \mathbf{0}$ has infinitely many solutions

Vector Space

n-space $\mathbf{R}^n$

$\mathbf{v}$

n-vector $\mathbf{v}$
 $(v_1, v_2, \dots, v_n)$

2-space $\mathbf{R}^2$

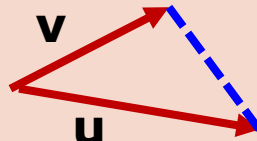
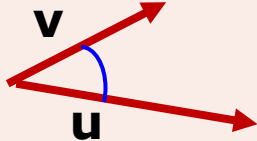
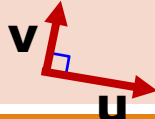
$(1,2)$ $(0.73, 2/5)$
 $(4,-3)$ (a,b)
...

3-space $\mathbf{R}^3$

$(1,2,3)$ $(0.73, 2/5, \sqrt{2})$
 $(4,-3, 0)$ (x,y,z)
...

Vector Operations and Relations

Let $\mathbf{u} = (u_1, u_2, \dots, u_n)$, $\mathbf{v} = (v_1, v_2, \dots, v_n)$ be n -vectors, and c a real number.

Vector addition	$\mathbf{u} + \mathbf{v}$	$(u_1+v_1, u_2+v_2, \dots, u_n+v_n)$
Scalar multiplication	$c \mathbf{v}$	$(cv_1, cv_2, \dots, cv_n)$
Dot product	$\mathbf{u} \cdot \mathbf{v}$	$u_1v_1 + u_2v_2 + \dots u_nv_n$
Norm (length)	$\ \mathbf{u}\ $	$\sqrt{\mathbf{u} \cdot \mathbf{u}}$
Distance		$\ \mathbf{u} - \mathbf{v}\ = \sqrt{(\mathbf{u} - \mathbf{v}) \cdot (\mathbf{u} - \mathbf{v})}$
Angle		$\cos^{-1} \left(\frac{\mathbf{u} \cdot \mathbf{v}}{\ \mathbf{u}\ \ \mathbf{v}\ } \right)$
Orthogonal		$\mathbf{u} \cdot \mathbf{v} = 0$

between
 $\mathbf{u}$ and $\mathbf{v}$

can involve more than 2 vectors
and for other dimensions

Linear combination

$$s(2, 1, 0) + t(-3, 0, 1)$$

Take $s = 1$, $t = 1$

$$1(2, 1, 0) + 1(-3, 0, 1) = (-1, 1, 1)$$

Take $s = 2$, $t = -1$

$$2(2, 1, 0) + (-1)(-3, 0, 1) = (7, 2, -1)$$

$(-1, 1, 1)$ and $(7, 2, -1)$
are called
linear combinations
of $(2, 1, 0)$ and $(-3, 0, 1)$

$(1, 1, 0)$ is **not a linear combination** of $(2, 1, 0)$ and $(-3, 0, 1)$

Cannot find s and t :

$$s(2, 1, 0) + t(-3, 0, 1) = (1, 1, 0)$$

Visualization

$$\mathbf{u}_1 = (2, 1, 3, 1), \mathbf{u}_2 = (1, -1, 2, 2), \mathbf{u}_3 = (3, 0, 5, 1)$$

Linear Combinations

$\mathbf{v}$ is a linear combination
of $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$

can find $a, b, c \rightarrow$

$$\mathbf{v} = a\mathbf{u}_1 + b\mathbf{u}_2 + c\mathbf{u}_3$$

$\mathbf{v}$ is not a linear combination
of $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$

$\leftarrow$ cannot find a, b, c

linear system has solution

$$\mathbf{v} = (3, 3, 4, 0)$$

$$(3, 3, 4, 0) = a(2, 1, 3, 1) + b(1, -1, 2, 2) + c(3, 0, 5, 1)$$

$$\begin{cases} 2a + b + 3c = 3 \\ a - b = 3 \\ 3a + 2b + 5c = 4 \\ a + 2b + c = 0 \end{cases}$$



$$\left(\begin{array}{ccc|c} 2 & 1 & 3 & 3 \\ 1 & -1 & 0 & 3 \\ 3 & 2 & 5 & 4 \\ 1 & 2 & 1 & 0 \end{array} \right) \xrightarrow{\text{G.E.}} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

$$\mathbf{u}_1 = (2, 1, 3, 1), \mathbf{u}_2 = (1, -1, 2, 2), \mathbf{u}_3 = (3, 0, 5, 1)$$

Linear Combinations

$\mathbf{v}$ is a linear combination
of $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$

can find $a, b, c \rightarrow$

$$\mathbf{v} = a\mathbf{u}_1 + b\mathbf{u}_2 + c\mathbf{u}_3$$

$\mathbf{v}$ is not a linear combination
of $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$

$\leftarrow$ cannot find a, b, c

linear system has no solution

$$(3, 3, 3, 0) = a(2, 1, 3, 1) + b(1, -1, 2, 2) + c(3, 0, 5, 1)$$

$$\begin{cases} 2a + b + 3c = 3 \\ a - b = 3 \\ 3a + 2b + 5c = 3 \\ a + 2b + c = 0 \end{cases}$$



$$\begin{pmatrix} 2 & 1 & 3 & 3 \\ 1 & -1 & 0 & 3 \\ 3 & 2 & 5 & 3 \\ 1 & 2 & 1 & 0 \end{pmatrix} \xrightarrow{\text{G.E.}} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Linear System (3 forms)

$$(3, 3, 4, 0) = a(2, 1, 3, 1) + b(1, -1, 2, 2) + c(3, 0, 5, 1)$$

$$\begin{cases} 2a + b + 3c = 3 \\ a - b = 3 \\ 3a + 2b + 5c = 4 \\ a + 2b + c = 0 \end{cases}$$

standard form

$$a \begin{pmatrix} 2 \\ 1 \\ 3 \\ 1 \end{pmatrix} + b \begin{pmatrix} 1 \\ -1 \\ 2 \\ 2 \end{pmatrix} + c \begin{pmatrix} 3 \\ 0 \\ 5 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \\ 4 \\ 0 \end{pmatrix}$$

vector equation form

$$\begin{pmatrix} 2 & 1 & 3 \\ 1 & -1 & 0 \\ 3 & 2 & 5 \\ 1 & 2 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \\ 4 \\ 0 \end{pmatrix}$$

matrix equation form

$\text{span}(S)$ is the set of all linear combinations of $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$

Can also be written: $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$

Linear Span

$S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ a finite collection of vectors in $\mathbf{R}^n$

$$\mathbf{R}^n = \text{span}(S)$$

$$c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \dots + c_k\mathbf{u}_k$$

$$\mathbf{u}_1 + 2\mathbf{u}_2 + \dots + 7\mathbf{u}_k$$

$$-3\mathbf{u}_1 + 0\mathbf{u}_2 + \dots + 2\mathbf{u}_k$$

S

$\mathbf{u}_1$

$\mathbf{u}_2$

$\dots$

$\mathbf{u}_k$

$$\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k \in \mathbf{R}^n$$

$$S \subseteq \mathbf{R}^n$$

$$\text{span}(S) \subseteq \mathbf{R}^n$$

$$S \subseteq \text{span}(S)$$

$$S \subseteq \text{span}(S) \subseteq \mathbf{R}^n$$

$\text{span}(S)$ can be equal to $\mathbf{R}^n$
but not always.

Example 3

standard basis vectors for $\mathbf{R}^2$

Which of the following vectors belong to $\text{span}\{ (1,0), (0,1) \}$?

Check whether each vector is a linear combination of $(1,0), (0,1)$.

- $(0,0)$
- $(1,0)$
- $(1,1)$
- $(1513, 2025)$

$$s(1,0) + t(0,1)$$

Are there vectors that do not belong to $\text{span}\{ (1,0), (0,1) \}$?

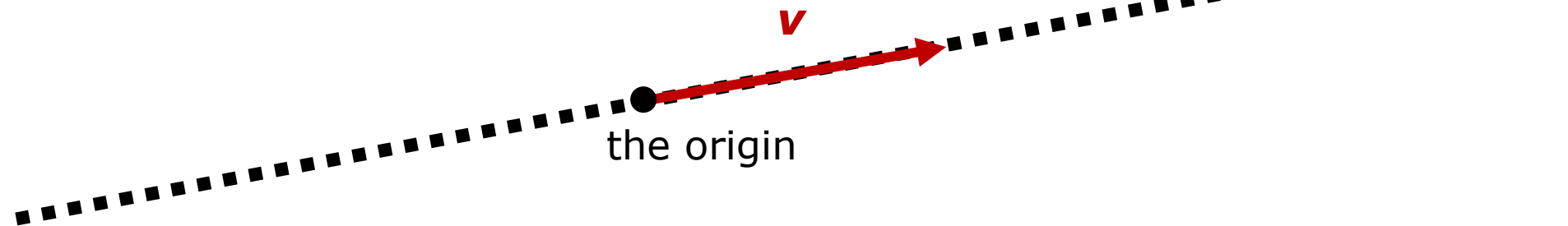
Linear Span of One Vector

Let $\mathbf{v} \in \mathbf{R}^n$ $\mathbf{v} \neq \mathbf{0}$

- $\text{span}\{\mathbf{v}\}$
- all scalar multiples $c\mathbf{v}$, $c \in \mathbf{R}$

In $\mathbf{R}^2$ and $\mathbf{R}^3$

- represent a line that passes through the origin and parallel to $\mathbf{v}$



Linear Span of Two Vectors

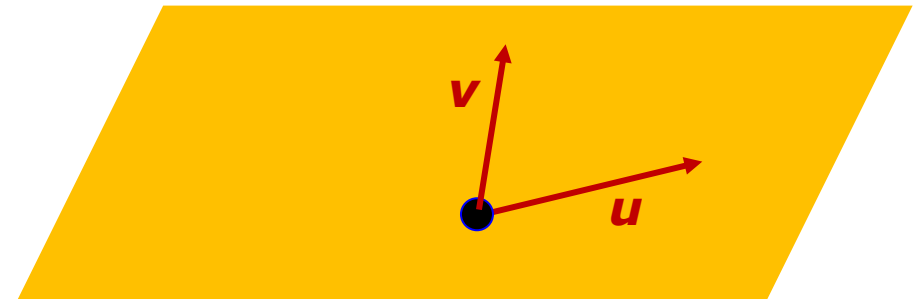
Let $\mathbf{u}, \mathbf{v} \in \mathbf{R}^n$ $\mathbf{u}, \mathbf{v} \neq \mathbf{0}$, not parallel to each other

- $\text{span}\{\mathbf{u}, \mathbf{v}\}$
- all linear combinations $c\mathbf{u} + d\mathbf{v}$, $c, d \in \mathbf{R}$

In $\mathbf{R}^3$

- Represent a plane that contains the origin and the vectors $\mathbf{u}, \mathbf{v}$

Visualization



Fill in the blanks

a vector in $\mathbb{R}^2$, a vector in $\mathbb{R}^3$, a line in $\mathbb{R}^3$, a plane in $\mathbb{R}^3$, the entire $\mathbb{R}^3$ space

1. A linear combination of two vectors in $\mathbb{R}^3$ is a vector in $\mathbb{R}^3$.
2. A linear combination of three vectors in $\mathbb{R}^3$ is a vector in $\mathbb{R}^3$.
3. A linear span of one non-zero vector in $\mathbb{R}^3$ is a line in $\mathbb{R}^3$.
4. A linear span of two non-parallel vectors in $\mathbb{R}^3$ is a plane in $\mathbb{R}^3$.
5. A linear span of three non-coplanar vectors in $\mathbb{R}^3$ is the entire $\mathbb{R}^3$ space.



Not quite right!